SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Collaborative Study Material & Quiz Platform (CSMQP)

Version: 1.0
Governing Standard: IEEE Std 830-1998

1. Introduction

1.1 Purpose

This document specifies the complete functional and non-functional requirements for the Collaborative Study Material & Quiz Platform (CSMQP). It serves as the primary technical specification for developers, quality assurance teams, and stakeholders.

1.2 Scope

CSMQP is a distributed full-stack web application designed for higher-education learning environments. The system supports:

- Peer-to-peer study material sharing (uploading, tagging, indexing, and downloading).
- Collaborative Question Bank curation and moderation.
- Synchronized multi-participant live quiz rooms and self-paced practice tests.
- Longitudinal student performance analysis with topic-level diagnostic feedback.

1.3 Definitions, Acronyms, and Abbreviations

- **SRS:** Software Requirements Specification
- **FR / NFR:** Functional Requirement / Non-Functional Requirement
- **RBAC:** Role-Based Access Control
- **JWT:** JSON Web Token
- **QB:** Question Bank
- **PA:** Performance Analysis
- **RTM:** Requirements Traceability Matrix

1.4 References

- IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications.

- ISO/IEC/IEEE 29148:2018: Systems and Software Engineering — Requirements Engineering.

1.5 Overview

Section 1 establishes context and definitions. Section 2 describes user roles, environment, constraints, and assumptions. Section 3 outlines verifiable functional, non-functional, interface, and security requirements. Section 4 provides the UML Use Case Structure, and Section 5 details the Requirements Traceability Matrix (RTM).

2. Overall Description

2.1 Product Perspective

CSMQP is a standalone 3-tier client-server application. A modern web frontend connects to containerized REST API and WebSocket services, backed by a relational PostgreSQL database and object storage.

2.2 User Classes and Characteristics

- **Student:** Can upload/download study materials, contribute multiple-choice questions, participate in live and practice quizzes, and view individual analytics dashboards.
- **Moderator / Instructor:** Validates questions, moderates flagged study notes, manages subject tags, and reviews aggregate cohort analytics.

2.3 Operating Environment

- **Client:** Modern web browsers (Chromium >= 100, Firefox >= 95, Safari >= 15) supporting WebSockets.
- **Backend:** Linux containers (Docker) running Node.js / Python FastAPI.
- **Database:** PostgreSQL 15+ and S3-compatible file storage.

2.4 Design and Implementation Constraints

- File uploads are capped at 25 MB per document.
- Network communication must use TLS 1.3 encryption.
- Relational access must strictly use parameterized queries to block SQL injection.

2.5 Assumptions and Dependencies

- Users possess broadband internet connections with latency <= 200 ms.
- Authentication tokens (JWT) have an expiration window of 60 minutes.

3. Specific Requirements

3.1 External Interface Requirements

- **User Interface:** Responsive Single Page Application (SPA) supporting desktop and mobile viewports.
- **Software Interface:** Integrates with PostgreSQL via TCP connection pooling and object storage via REST APIs.
- **Communication Interface:** HTTPS on port 443; WSS (WebSocket Secure) for real-time quiz synchronization.

3.2 Functional Requirements (FRs)

- **FR-01 (Authentication & Authorization):** The system shall permit users to register and log in via email and password, issuing a signed JWT with user identity and RBAC role upon successful validation.
- **FR-02 (Study Material Sharing):** The system shall allow authenticated students to upload study notes (PDF, Markdown) up to 25 MB, associating them with a title, course code, and comma-separated tags, and allow users to search and download them.
- **FR-03 (Collaborative Question Bank):** The system shall allow users to contribute multiple-choice questions (prompt, 2-5 choices, correct answer key, and explanation), tag them by course code and topic, and search verified questions.
- **FR-04 (Synchronized & Practice Quizzes):** The system shall allow a host to create a quiz room from selected Question Bank items with a unique 6-character room code, and synchronize question timers and answer states across all connected peers in real time.
- **FR-05 (Performance Analysis):** The system shall track individual attempt logs, computing topic-wise accuracy percentages, average time-per-question, and cohort percentile ranks, rendering these metrics on an interactive dashboard.

3.3 Non-Functional Requirements (NFRs)

- **NFR-01 (Analytics Query Latency):** The system shall compute and return historical performance dashboard data in under 600 ms for accounts with up to 1,000 recorded question attempts.
- **NFR-02 (Real-Time Broadcast Latency):** The system shall propagate active question shifts and timer state events to all room participants in under 150 ms.
- **NFR-03 (System Availability):** The platform shall maintain high operational availability outside scheduled maintenance windows.

3.4 Security Objectives & Requirements

- **Security Objective 1 (Data Privacy & Confidentiality):** Prevent unauthorized access to private student analytics and sensitive account credentials.

    - **SEC-01:** The system shall hash user passwords using Bcrypt with a work factor >= 12 and enforce strict RBAC checks to ensure students can only view or modify their own data.
- **Security Objective 2 (Quiz Integrity & Anti-Tampering):** Prevent participants from intercepting answer keys before question evaluation.
    - **SEC-02:** The system shall withhold correct answer payloads and explanations on the server until the active timer has expired or all participants have submitted answers.

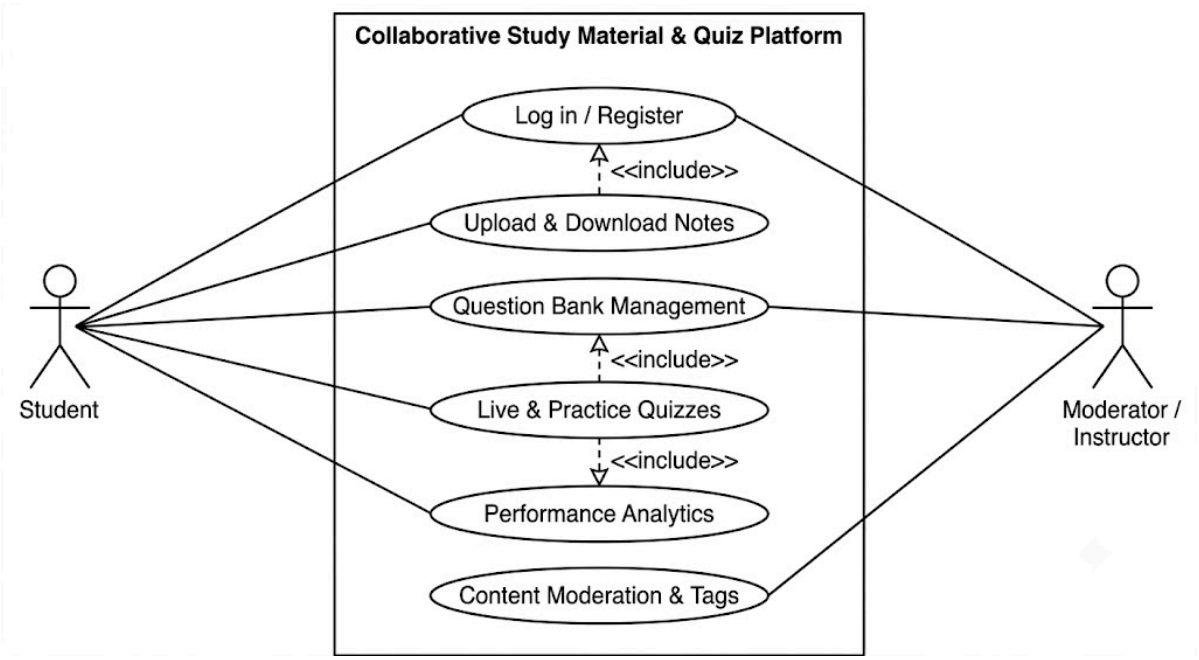
4. UML Use Case Structure

Primary Actors:

1. **Student:** Primary consumer and contributor of notes, quizzes, and personal analytics.
2. **Moderator / Instructor:** Content curator and supervisor of question quality and course tags.

Use Case Mappings:

- **UC-01 (Login / Register):** Realizes FR-01. Mandatory prerequisite for note uploads and hosting quizzes.
- **UC-02 (Upload & Search Study Notes):** Realizes FR-02. Includes authentication.
- **UC-03 (Browse & Contribute Question Bank):** Realizes FR-03. Allows adding and filtering MCQs.
- **UC-04 (Create & Host Live Quiz):** Realizes FR-04. Includes Question Bank selection.
- **UC-05 (Join & Play Quiz):** Realizes FR-04. Includes answer submission and score calculation.
- **UC-06 (View Performance Analytics):** Realizes FR-05. Shows diagnostic breakdowns and trends.
- **UC-07 (Moderate Content):** Restricted to Moderator. Reviewing flagged materials and question entries.
- **UC-08 (Manage Subject Tags):** Restricted to Moderator. Managing global course and subject taxonomy.


Collaborative Study Material & Quiz Platform
Log in / Register
<<include>>
Upload & Download Notes
Question Bank Management
<<include>>
Live & Practice Quizzes
<<include>>
Performance Analytics
Content Moderation & Tags
Student
Moderator / Instructor

5. Requirements Traceability Matrix (RTM)

Requirement ID	Category	Summary Description	Architecture Component	Verification Method	Test Case ID
FR-01	Functional	User authentication & JWT session management	Auth Middleware / AuthSvc	Automated API Test	TC-01, TC-02
FR-02	Functional	Study material upload (PDF <= 25MB) & search	Material Service & DocStore	Boundary & Form Test	TC-03, TC-04
FR-03	Functional	Collaborative Question Bank CRUD & tagging	Question Bank Service	Query & CRUD Test	TC-05
FR-04	Functional	Live quiz room synchronization & scoring	WebSocket Quiz Engine	Multi-Client Integration	TC-06
FR-05	Functional	Student performance analysis & trend dashboards	Performance Analytics Engi...	Data Aggregation Test	TC-07, TC-08
NFR-01	Non-Functional	Analytics query response time <= 600 ms	Analytics Engine / DB Index...	Benchmark (k6/Locust)	TC-08
NFR-02	Non-Functional	WebSocket live state broadcast <= 150 ms	WebSocket Quiz Engine	Network Latency Test	TC-06
NFR-03	Non-Functional	99.5% service uptime & persistent logging	Docker Engine / DB WAL	Audit & Load Testing	
SEC-01	Security	Bcrypt password hashing & RBAC privacy	Auth Middleware / DB	Security / IDOR Audit	
SEC-02	Security	Server-side withholding of quiz answer keys	WebSocket Quiz Engine	Payload Inspection	TC-10